Optimization of Genomic Classifiers for Clinical Deployment: Evaluation of Bayesian Optimization to Select Predictive Models of Acute Infection and In-Hospital Mortality

Inflammatix
Advanced host-response diagnostics

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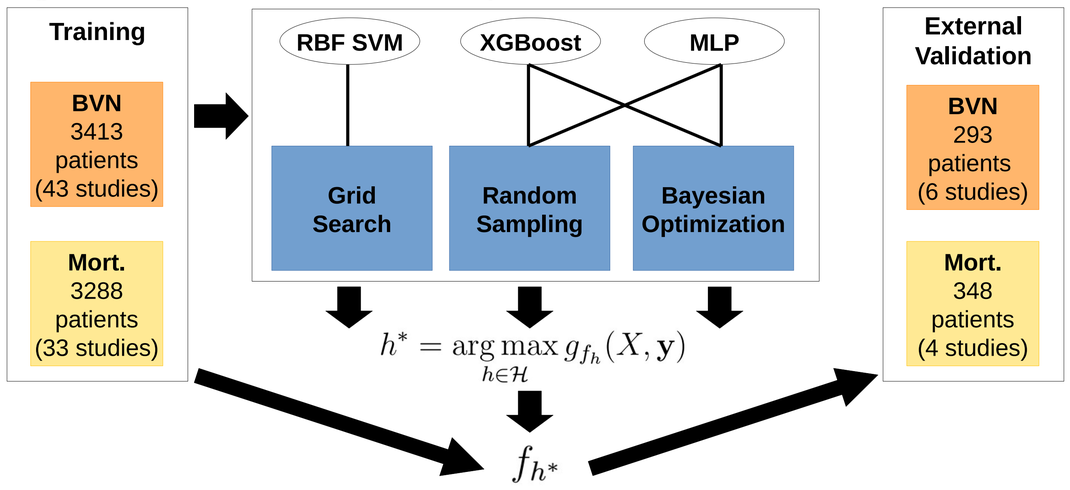
Background & Motivation

- Patient severity of illness and infection status are difficult to predict rapidly and accurately using conventional diagnostics (e.g. physiological symptoms, lab cultures)
- Quantifying the *host response* via measurement of the gene expression of specific diagnostic markers from patient blood has proven effective for these two tasks[1,2]
- Development and deployment of diagnostic classifiers depends on hyperparameter optimization (HO): tuning of elements of the classifier's training or structure
- Numerous HO approaches, including grid search (GS), random sampling[3] (RS) and Bayesian optimization[4] (BO), have proven successful but have not been compared for molecular diagnostics
- We evaluate performance (in both internal and external validation) of gene expression-based diagnostic classifiers selected by GS, RS and BO for multi-class prediction of acute infection (bacterial-viral-noninfected; BVN) and prediction of 30-day mortality

Comparing Hyperparameter Optimization Approaches for Diagnostic Classifier Development



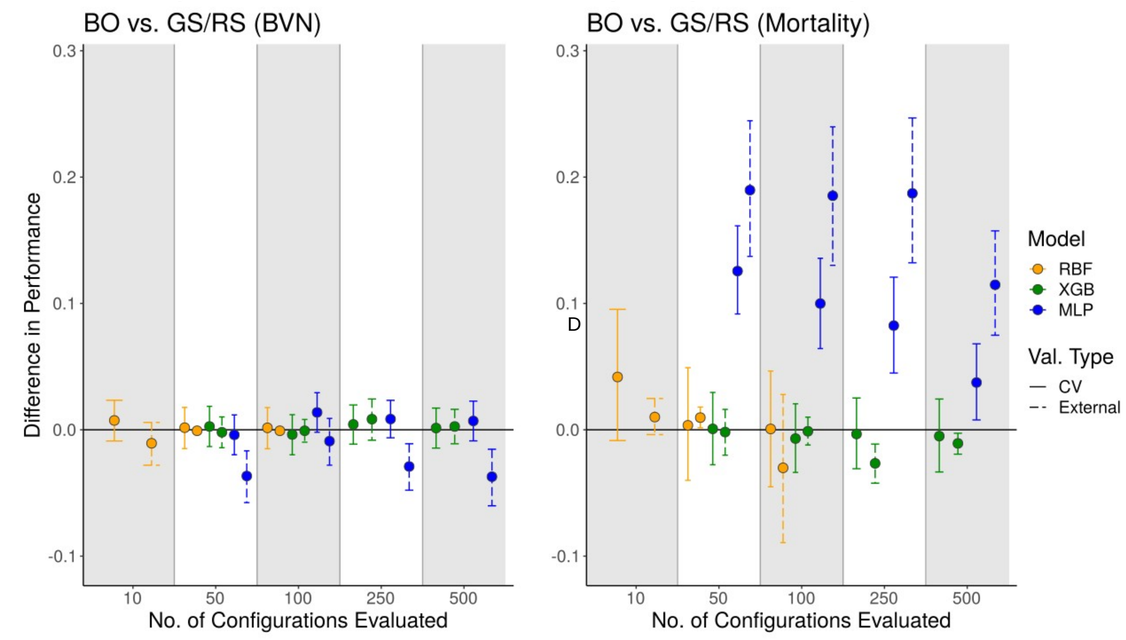
Hyperparameter configurations (*h*) were evaluated for each classifier type by grouped five-fold cross-validation (*g*) using gene expression from 29 diagnostic markers (*X*) and the task-relevant class labels (***y***). Optimal classifiers (f_{h^*}) for each HO setting and classifier type were trained and applied on the corresponding validation sets. Further details of BO settings considered in [5].

Performance Overview of Classifiers Selected by Grid Search/Random Sampling or Bayesian Optimization

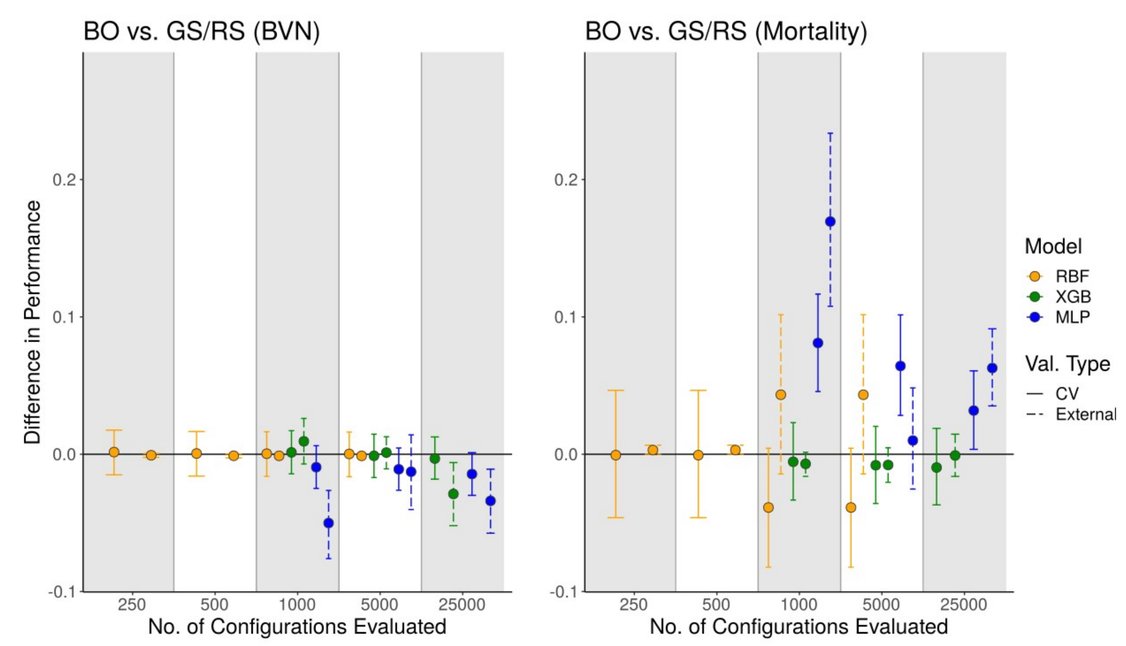
Model	HO Type	No. of Evals.	ARD	BVN CV	BVN Val.	Mortality CV	Mortality Val.
RBF	GS	10	-	0.808	0.862	0.758 (0.182)	0.736 (0.375)
	GS	50	-	0.814	0.853	0.797 (0.169)	0.739 (0.372)
	GS	100	-	0.814	0.853	0.800 (0.192)	0.782 (0.533)
	GS	250	-	0.814	0.853	0.801 (0.191)	0.749 (0.386)
	GS	500	-	0.815	0.853	0.801 (0.191)	0.749 (0.386)
	GS	1000	-	0.815	0.853	0.839 (0.225)	0.708 (0.444)
	GS	5000*	-	0.815	0.853	0.839 (0.225)	0.708 (0.444)
	BO	10	Y	0.811	0.788	0.800 (0.190)	0.747 (0.383)
	BO	10	N	0.815	0.851	0.800 (0.187)	0.746 (0.381)
	BO	50	Y	0.816	0.852	0.801 (0.196)	0.752 (0.389)
	BO	50	N	0.816	0.852	0.801 (0.194)	0.749 (0.385)
	BO	100	Y	0.816	0.852	0.800 (0.197)	0.753 (0.392)
	BO	100	N	0.816	0.852	0.801 (0.196)	0.752 (0.389)
XGB	RS	50	-	0.809	0.830	0.880 (0.315)	0.819 (0.542)
	RS	100	-	0.813	0.827	0.885 (0.288)	0.819 (0.526)
	RS	250	-	0.812	0.826	0.885 (0.308)	0.829 (0.556)
	RS	500	-	0.810	0.829	0.885 (0.320)	0.826 (0.559)
	RS	1000	-	0.810	0.822	0.885 (0.311)	0.822 (0.552)
	RS	5000	-	0.813	0.830	0.888 (0.310)	0.823 (0.552)
	RS	25000	-	0.815	0.860	0.889 (0.303)	0.816 (0.532)
	BO	50	Y	0.818	0.865	0.887 (0.301)	0.814 (0.540)
	BO	50	N	0.812	0.828	0.881 (0.275)	0.817 (0.516)
	BO	100	Y	0.811	0.825	0.885 (0.314)	0.825 (0.559)
	BO	100	N	0.809	0.826	0.878 (0.288)	0.817 (0.521)
	BO	250	Y	0.818	0.865	0.886 (0.290)	0.826 (0.539)
	BO	250	N	0.816	0.834	0.882 (0.272)	0.802 (0.483)
	BO	500	Y	0.818	0.865	0.889 (0.346)	0.827 (0.591)
	BO	500	N	0.812	0.831	0.880 (0.313)	0.815 (0.538)
MLP	RS	50	-	0.818	0.860	0.763 (0.121)	0.631 (0.288)
	RS	100	-	0.814	0.863	0.785 (0.156)	0.640 (0.301)
	RS	250	-	0.824	0.861	0.807 (0.211)	0.625 (0.366)
	RS	500	-	0.819	0.859	0.853 (0.240)	0.691 (0.401)
	RS	1000	-	0.835	**0.872**	0.809 (0.158)	0.637 (0.333)
	RS	5000	-	0.837	0.835	0.826 (0.249)	0.796 (0.546)
	RS	25000	-	**0.840**	0.856	0.859 (0.267)	0.743 (0.428)
	BO	50	Y	0.816	0.820	0.888 (0.340)	0.823 (0.554)
	BO	50	N	0.814	0.824	0.888 (0.290)	0.820 (0.564)
	BO	100	Y	0.822	0.845	0.886 (0.296)	**0.847** (0.631)
	BO	100	N	0.828	0.854	0.884 (0.292)	0.825 (0.577)
	BO	250	Y	0.817	0.848	0.890 (0.312)	0.842 (0.614)
	BO	250	N	0.832	0.832	0.889 (0.335)	0.812 (0.566)
	BO	500	Y	0.837	0.855	**0.894** (0.304)	0.835 (0.593)
	BO	500	N	0.826	0.822	0.890 (0.330)	0.806 (0.561)

Performance in internal (CV) and external validation for all three classifier types at all hyperparameter optimization settings. MLP classifiers generally demonstrated the highest performance across both tasks. BVN performance is measured by Hand & Till's multi-class AUC and 30-day mortality performance is measured by binary AUC.

Classifiers Selected by Grid/Random Search Compare Favorably with those Selected by Bayesian Optimization

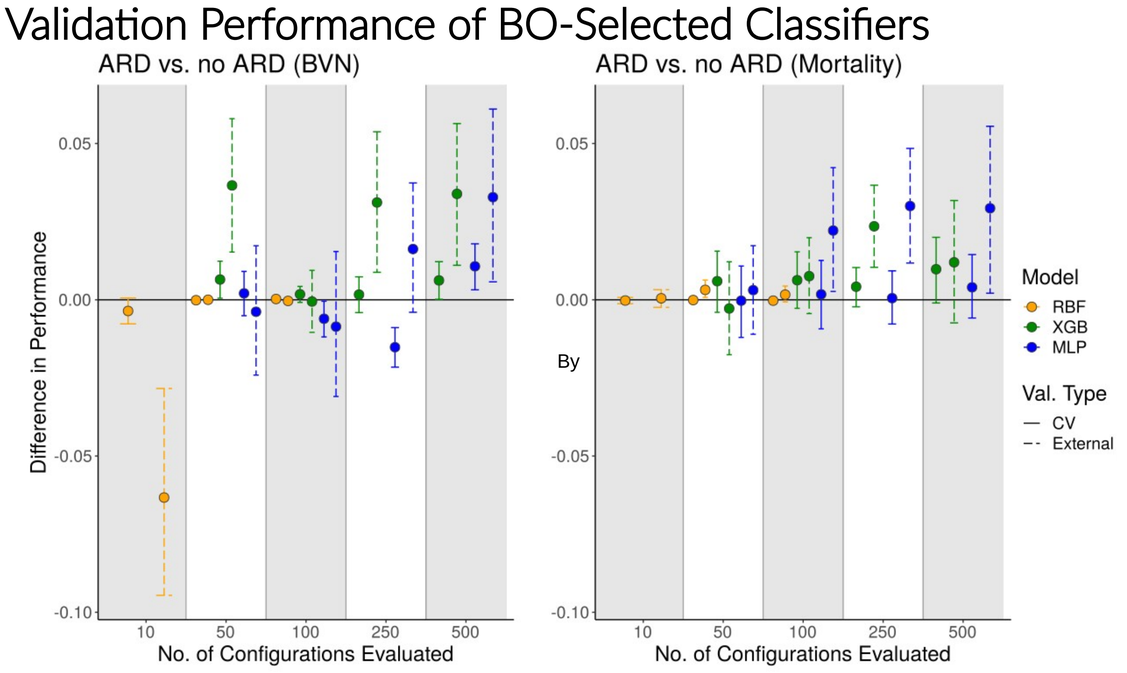


Comparison of performance between classifiers selected by BO or GS/RS at smaller numbers of evaluated hyperparameter configurations (more typical of BO). Differences in performance greater than 0 reflect higher performance by the BO-selected classifier. BO-selected MLP classifiers for the mortality task show higher performance than their GS/RS-selected counterparts.



At larger numbers of evaluated hyperparameter configurations (more typical of GS/RS), classifiers selected by either GS/RS or BO still show comparable performance (again, with the notable exception of BO-selected MLP classifiers of 30-day mortality).

Automatic Relevance Determination has Limited Impact on Validation Performance of BO-Selected Classifiers



Comparison of performance between classifiers selected by BO with ARD enabled or disabled. Though ARD-enabled BO did result in classifiers with slightly higher external validation performance for both tasks, changes in the BO acquisition function or search space resulted in no such differences (see Mayhew et al. Supplement[5])

Conclusion

- MLPs demonstrated the highest performance for both classification tasks, regardless of the HO approach used
- Classifiers selected by GS/RS were comparable in performance with those selected by BO even at smaller computational budgets typical of BO
- However, BO-selected MLP classifiers of 30-day mortality remained higher performing even at larger computational budgets compared to their GS/RS-selected counterparts
- A conventional setting in Gaussian process-based BO (automatic relevance determination) provided marginal gains in performance of selected classifiers
- Both GS/RS and BO remain promising avenues for hyperparameter optimization and represent key components in the development of more effective diagnostics for emergency and critical care

References

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